

HDOC – Day 22

C Programming Assignment

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Date - 15/09/2026

Problem: Accept two positive integers m and n where $m \leq n$. Repeatedly display a menu and use a switch statement to perform multiplication tables, a multiplication grid, consecutive-integer output, or exit. Invalid menu choices must produce an error message.

Algorithm

1. Start.
2. Read positive integers m and n .
3. If $m \leq 0$, $n \leq 0$, or $m > n$, display an invalid-input message and stop.
4. Use a do-while loop to display the menu until Exit is selected.
5. Read the menu choice and use a switch statement.
6. Choice 1: for $i = 1$ to 10, print $m \times i$ and $n \times i$ side by side.
7. Choice 2: print column headings 1 to n ; for each row $i = 1$ to m , print $i \times j$ for $j = 1$ to n .
8. Choice 3: for each i from m to n , print i followed by $i+1$, $i+2$, $i+3$, $i+4$, and $i+5$.
9. Choice 4: display the exit message and terminate the loop.
10. Default: display an invalid menu choice message, then show the menu again.
11. Stop.

Test-Case Table

TC	m	n	Choice	Expected Result	Status
1	3	5	1	Tables of 3 and 5 from multiplier 1 to 10	Pass
2	3	5	2	3 x 5 multiplication grid	Pass
3	3	5	3	Rows for 3, 4, 5; each followed by next five integers	Pass
4	3	5	7	Invalid menu choice message; menu shown again	Pass
5	0	5	-	Invalid input message because m is not positive	Pass

C Program

```
#include <stdio.h>
int main(void) {
    int m, n, choice, i, j;
    printf("Enter two positive integers m and n (m <= n): ");
    scanf("%d %d", &m, &n);
    if (m <= 0 || n <= 0 || m > n) {
        printf("Invalid input. Please enter positive integers with m <= n.\n");
        return 0;
    }
    do {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n", m, n, m, n);
                for (i = 1; i <= 10; i++)
                    printf("%d x %d = %-4d %d x %d = %d\n", m, i, m*i, n, i, n*i);
                break;
```

case 2:

```
printf("\nMultiplication Grid (m=%d, n=%d)\n", m, n);
```

```
printf(" ");
```

```
for (j = 1; j <= n; j++) printf("%4d", j);
```

```
printf("\n ");
```

```
for (j = 1; j <= n; j++) printf("----");
```

```
printf("\n");
```

```
for (i = 1; i <= m; i++) {
```

```
printf("%2d |", i);
```

```
for (j = 1; j <= n; j++) printf("%4d", i*j);
```

```
printf("\n");
```

```
}
```

```
break;
```

case 3:

```
printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n", m, n, m, n);
```

```
for (i = m; i <= n; i++) {
```

```
printf("%d:", i);
```

```
for (j = 0; j <= 5; j++) printf(" %d", i+j);
```

```
printf("\n");
```

```
}
```

```
break;
```

case 4:

```
printf("Exiting program.\n");
```

```
break;
```

default:

```
printf("Invalid menu choice. Please select 1, 2, 3, or 4.\n");
```

```
}
```

```
} while (choice != 4);
```

```
return 0;
```

```
}
```

Compilation, Execution and Testing

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ vi pro.c

(kali㉿kali)-[~]
$ cc pro.c

(kali㉿kali)-[~]
$ ./a.out
Enter two positive integers m and n ( $m \leq n$ ): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5 (m=3, n=5)
3 x 1 = 3    5 x 1 = 5
3 x 2 = 6    5 x 2 = 10
3 x 3 = 9    5 x 3 = 15
3 x 4 = 12   5 x 4 = 20
3 x 5 = 15   5 x 5 = 25
3 x 6 = 18   5 x 6 = 30
3 x 7 = 21   5 x 7 = 35
3 x 8 = 24   5 x 8 = 40
3 x 9 = 27   5 x 9 = 45
3 x 10 = 30  5 x 10 = 50
```

```
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

Multiplication Grid (m=3, n=5)
  1  2  3  4  5
  --
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 7
Invalid menu choice. Please select 1, 2, 3, or 4.
```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 4

Exiting program.

`(kali@kali)-[~]`

`$ ./a.out`

Enter two positive integers m and n ($m \leq n$): 0 5

Invalid input. Please enter positive integers with $m \leq n$.

`(kali@kali)-[~]`